

MacBook Feasibility Experiment Plan

Trajectory-Derived Counterfactual Action Landscapes for PPO

Goal: verify the complete pipeline on MountainCar before scaling. This is a feasibility/mechanism test, not the final paper experiment.

Pipeline: PPO trajectories → explicit counterfactual reference → COCOA landscape → state→landscape model $B(s)$ → PPO regularization → stability/performance checks.

Notebook	Purpose	Minimal run
01_train_ppo_baseline	Working PPO + trajectory/checkpoint generation	MountainCar; 3 seeds; 50k–100k steps/seed
02_counterfactual_oracle	Explicit all-action reference landscape	500 states; 3 actions/state; ~30% checkpoint
03_cocoa_landscape	Trajectory-only counterfactual landscape	Same trajectories/states; no oracle labels
04_validate_landscape	COCOA vs explicit reference	MSE, correlation, best-action agreement
05_state_to_landscape	Learn $B(s) \approx A_{CF}(s, \cdot)$	70/30 split; Ridge + tiny MLP
06_cf_ppo_smoke_test	Test landscape-based PPO regularization	PPO vs CF-PPO vs shuffled; 3 seeds; $\lambda=0.1$

01 — train_ppo_baseline.ipynb

Train standard PPO and create the reusable trajectory/checkpoint dataset.

- Inputs: MountainCar-v0, fixed PPO config, 3 seeds.
- Save observations, actions, rewards, next states, done flags, log-probs, policy probabilities, values, episode/timestep IDs.
- Save checkpoints at ~10%, 30%, 50%, 75%, 100%.
- Outputs: return curve, entropy, KL, losses, trajectories, checkpoints.
- **CHECK:** PPO learns at least partially; data and checkpoints reload correctly.

02 — counterfactual_oracle.ipynb

Create a small simulator-based reference landscape.

- Use 500 states from the ~30% checkpoint.
- For every state, restore simulator state and execute actions 0, 1, 2.
- Compute $Q_{CF}(s,a)=r+\gamma V(s')$; terminal $V=0$.
- Compute $V_{\pi}(s)=\sum \pi(a|s)Q_{CF}(s,a)$, then $A_{CF}=Q_{CF}-V_{\pi}$.
- **CHECK:** $\sum \pi A \approx 0$ per state; action values differ on a meaningful subset; restore/step logic is correct.

03 — cocoa_landscape.ipynb

Infer the counterfactual action landscape using only ordinary PPO trajectories.

- Input only the trajectories from Notebook 01.
- Run/adapt COCOA to produce an all-action counterfactual/contribution signal.
- Do not provide explicit counterfactual labels.
- Save $A_{COCOA}(s,\cdot)$ for the same evaluation states.
- **CHECK:** no extra counterfactual simulator calls are required for estimation; a K-dimensional signal is produced for each state.

04 — validate_landscape.ipynb

Test whether COCOA recovers the explicit action landscape.

- Compare A_{COCOA} with A_{CF} on identical states.
- Metrics: vector MSE, Pearson/Spearman correlation, best-action agreement, magnitude ratio.
- Make per-action scatter plots and several individual-state comparisons.
- Also compare policy-gradient directions if feasible.
- **CHECK:** meaningful positive agreement. If COCOA is near random, stop and debug before proceeding.

05 — state_to_landscape.ipynb

Test the amortized state→action-landscape model.

- Dataset: $(s, A_{COCOA}(s,\cdot))$; 70% train / 30% held-out.
- Models: Ridge first; tiny MLP second.
- Metrics: R^2 per action, vector MSE, best-action agreement.
- Critical: evaluate $B(s)$ against explicit A_{CF} on held-out states.
- Control: shuffle state-target pairs; predictive performance should collapse.
- **CHECK:** $B(s)$ generalizes and retains agreement with explicit CF.

06 — cf_ppo_smoke_test.ipynb

Test whether the learned landscape can regularize PPO.

- Conditions: standard PPO; PPO + $\lambda B(s)$; PPO + shuffled $B(s)$.
- Start with $\lambda=0.1$ only; 3 seeds.
- Use $A_{\text{new}}=A_{\text{GAE}}+\lambda B(s)$, with consistent centering/scaling of B .
- Log return, environment steps, entropy, KL, gradient norm, and directional alignment.
- **CHECK:** B changes updates coherently; shuffled B does not reproduce the same effect; no catastrophic instability.

Go / No-Go Gates

Gate	Must observe before continuing
1. PPO	Baseline learns and trajectory files are valid.
2. Explicit CF	A_CF is nontrivial, numerically stable, and policy-centered.
3. COCOA	Trajectory-only estimator produces an all-action signal.
4. Landscape agreement	COCOA has meaningful agreement with explicit CF.
5. Amortization	B(s) predicts held-out landscapes and agrees with explicit CF.
6. PPO intervention	CF-PPO changes update direction/stability coherently; shuffled control does not replicate it.

Minimum outputs to inspect

1. PPO return vs environment steps.
2. Explicit A_CF distributions and per-state advantage spread.
3. Explicit vs COCOA scatter plots for each action.
4. COCOA vs explicit best-action agreement.
5. Held-out R²/MSE for B(s), plus B(s) vs explicit CF.
6. PPO vs CF-PPO vs shuffled return curves.
7. Gradient alignment with the counterfactual reference.
8. KL per update and policy entropy.

Do not implement yet

Skip LunarLander/Pendulum, large seed counts, hyperparameter sweeps, online B updates, adaptive λ , directional rotation, complex triggers, and paper-quality statistics. Those come only after the six-notebook pipeline passes.

Feasibility criterion

The first pass succeeds if the chain **trajectory** → **COCOA** → **action landscape** → **B(s)** → **PPO intervention** executes and the intermediate signals are coherent. A performance improvement is desirable but is not required for this first debugging pass.